Quantifying Seasonal Water Volume Fluctuations in Asia's Largest Brackish-Water Lagoon: A Remote Sensing–GIS Approach for Chilika, India (2023–2025)

Methodology

1. 1. Study Area

This study focuses on Chilika Lagoon, the largest brackish-water lagoon in India, located along the east coast in the state of Odisha. The lagoon experiences strong seasonal hydrological variability driven by monsoon rainfall, freshwater inflow from surrounding rivers, and tidal exchange with the Bay of Bengal, with published estimates placing its water-spread area at approximately 900 km² during the dry season and up to 1,165 km² during the monsoon (Samal, 2011). These seasonal hydrodynamic processes govern the lagoon's water-spread area, bathymetry, and total water volume. For this study, lagoon volume was estimated seasonally — summer, monsoon, and winter — for 2023–2025, using field-based bathymetric depth observations combined with satellite-derived water extent.

2. Data Sources

2.1 Bathymetric Depth Data. Point-based depth measurements were collected during field surveys between 2023 and 2025, and converted to positive depth-below-surface values for consistency in volume calculations. These points capture the spatial variability of lagoon-bottom morphology across different sectors, which is itself considerable: published bathymetric surveys show the northern sector as shallowest (0.5–1.5 m), the central sector at moderate depth (1–2.5 m), and the southern sector reaching a maximum of around 3.9 m.

2.2 Satellite Imagery. Seasonal water-spread area was extracted from Landsat 8 and Landsat 9 imagery (Roy et al., 2014). Surface water was delineated using the Modified Normalized Difference Water Index (MNDWI; Xu, 2006), which enhances open-water features while suppressing built-up land and vegetation. Cloud-free scenes representing summer (March–May), monsoon (June–September), and winter (October–February) were selected for each year.

3. Extraction of Seasonal Water Spread Area

Water bodies were delineated using MNDWI, calculated as:

$$\text{MNDWI} = (\text{Green} - \text{SWIR}) / (\text{Green} + \text{SWIR})$$

where Green is reflectance in the green band and SWIR is reflectance in the shortwave-infrared band. Pixels with positive MNDWI values were classified as water; threshold-based classification isolated lagoon water from surrounding land. The resulting water mask was converted to a vector polygon representing seasonal water-spread area, which defined the spatial extent used for volume estimation.

4. Preparation of the Bathymetric Surface

Depth values were verified and converted to positive depth-below-surface measurements, then spatially interpolated using a Triangulated Irregular Network (TIN) in ArcGIS Pro, representing bathymetric variation as a network of triangular facets derived from the irregularly distributed depth points. The TIN was subsequently converted to a raster-based Digital Bathymetric Model (DBM) at a consistent spatial resolution.

5. Seasonal Volume Estimation

Lagoon volume was estimated separately for each season and year (2023–2025) as follows: (i) the MNDWI-derived seasonal water polygon was prepared; (ii) the bathymetric raster was clipped to that polygon so calculations included only submerged area; and (iii) volume was computed by integrating depth across raster cells:

$$\mathbf{V = \Sigma(D_i \times A)}$$

where V is total water volume, D_i is the depth value of raster cell i, and A is the area of one raster cell. Summing depth–area products across all cells within the seasonal water extent yielded total lagoon volume for that season.

[Note: I've removed the separate Kriging-based grid-cell workflow (formerly Sections 5–7) pending your confirmation of which method actually produced the results below — see flag above.]

6. Results and Discussion

Year	Season	Area in sq. Meter	Volume in MCM
2023	Summer	790.88	1406.38
	Monsoon	871.01	1595.97
	Winter	841.03	1588.30
2024	Summer	738.24	1665.44
	Monsoon	860.31	1748.38
	Winter	842.22	1965.28
2025	Summer	776.40	1549.11
	Monsoon	851.61	2280.94
	Winter	842.16	1338.81

6.1 Seasonal Variation in Lagoon Area and Volume

The seasonal analysis of water-spread area and volume for 2023–2025 revealed marked temporal variability tied to the lagoon's hydrological cycle. Computed area ranged from 738.24 to 871.01 km², while volume ranged from 1,338.81 to 2,280.94 MCM. It is worth noting that these area estimates run somewhat below the commonly cited range for Chilika (approximately 900–1,165 km²; Samal, 2011); this gap likely reflects differences in MNDWI threshold sensitivity near turbid, shallow lagoon margins, or the specific scene dates selected, and should be discussed as a source of uncertainty rather than treated as a direct contradiction of prior estimates.

6.1.1 Summer. Summer showed comparatively lower water-spread area and moderate volume. In 2023, the lagoon covered 790.88 km² with an estimated volume of 1,406.38 MCM. A reduction in area to 738.24 km² occurred in 2024, though volume rose to 1,665.44 MCM — indicating a shift in depth distribution rather than a simple area–volume relationship. In 2025, area recovered slightly to 776.40 km², with volume of 1,549.11 MCM. The smaller summer extent is consistent with limited freshwater inflow, elevated evaporation, and reduced precipitation, which expose shallow peripheral zones and reduce overall water spread — a pattern consistent with the lagoon's known seasonal contraction (Samal, 2011).

6.1.2 Monsoon. Monsoon recorded the largest extent and, in most years, the highest volume, reflecting the dominant influence of rainfall and river discharge. Area reached 871.01 km² in 2023 (volume: 1,595.97 MCM) and 860.31 km² in 2024 (volume: 1,748.38 MCM). In 2025, volume rose sharply to 2,280.94 MCM despite a marginally smaller area (851.61 km²), suggesting greater depth accumulation from elevated freshwater inflow or hydrodynamic retention within the basin that year — though this single-year spike is large enough that it would be worth cross-checking against rainfall records for 2025 before presenting it as a robust trend.

6.1.3 Winter. Winter areas were comparatively stable relative to summer: 841.03 km² (2023, volume 1,588.30 MCM) and 842.22 km² (2024, volume 1,965.28 MCM, indicating continued deep-water retention following the monsoon). In 2025, however, area held nearly constant (842.16 km²) while volume fell sharply to 1,338.81 MCM — the lowest value in the dataset despite unchanged area, pointing to post-monsoon drainage, tidal flushing, or a change in the interpolated depth surface for that period specifically.

6.2 Interannual Variability (2023–2025)

Lagoon volume varied considerably across the study period, from a maximum of 2,280.94 MCM (monsoon 2025) to a minimum of 1,338.81 MCM (winter 2025) — the same year producing both the highest and lowest recorded volumes, which underscores how strongly the lagoon's storage capacity responds to within-year hydrological swings rather than gradual multi-year trends. These fluctuations likely reflect variation in annual rainfall, river discharge, sedimentation, and lagoon–sea exchange.

6.3 Hydrodynamic Implications

The seasonal expansion in area and volume during monsoon reflects the lagoon's function as a natural reservoir for freshwater storage and flood buffering, consistent with its broader role as a Ramsar-listed wetland supporting fisheries and biodiversity (Ramsar Convention Secretariat, 2022). The summer contraction reflects evaporation, reduced inflow, and tidal flushing, while the relative stability of winter extent suggests a transitional, moderately balanced hydrological state between the two extremes. Combining satellite-derived water extent with bathymetric interpolation offers a practical, repeatable approach to tracking the lagoon's seasonal water-storage dynamics, though the area discrepancy noted above suggests the MNDWI threshold and reference water-line definition used here would benefit from validation against an independent boundary (e.g., CDA field surveys) in future work.

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